

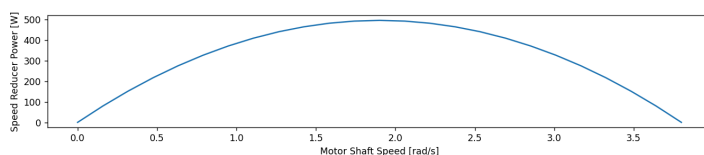
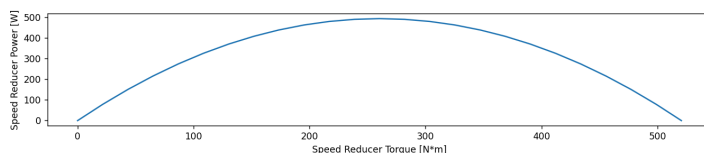
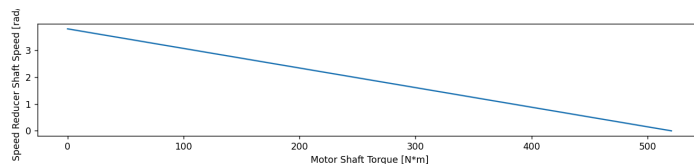
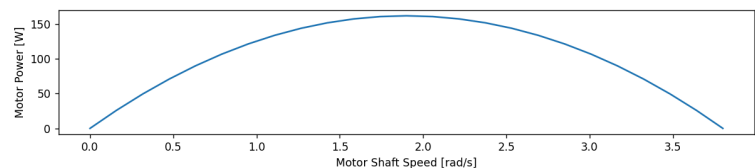
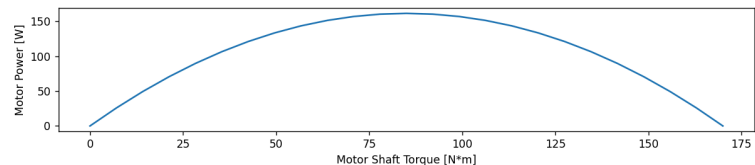
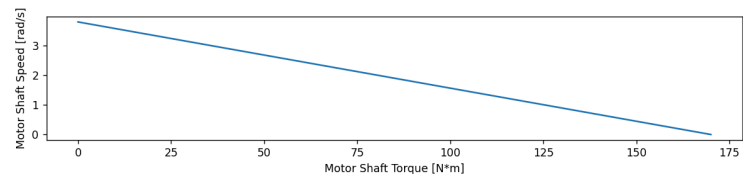
Coding

1. There is an advantage to defining gravitational acceleration in an outside structure rather than directly inputting the value into the functions. Keeping gravity as a variable allows us to examine its effect on the function outputs easily, while also making the code easier to understand.
2. Calling `F_gravity` using a terrain slope of 115 degrees raises an exception (Argument 'terrain_angle' must have values between -75 and +75 degrees). The input value for terrain slope exceeds a reasonable value, which is caught by a line in `F_gravity` where it detects that the value of the slope is greater than 75 degrees, causing the error. This is desirable behavior since it keeps unreasonable values (like 115 degrees) out of our calculations.

Motor and Speed Reducer

Behavior

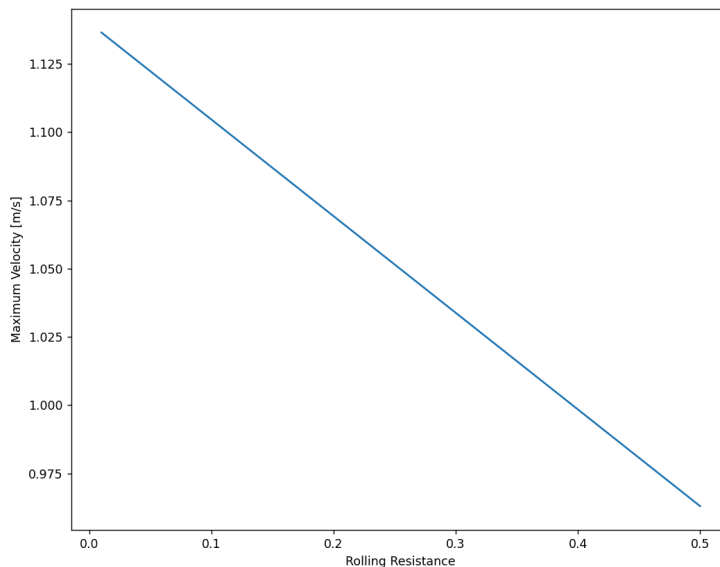
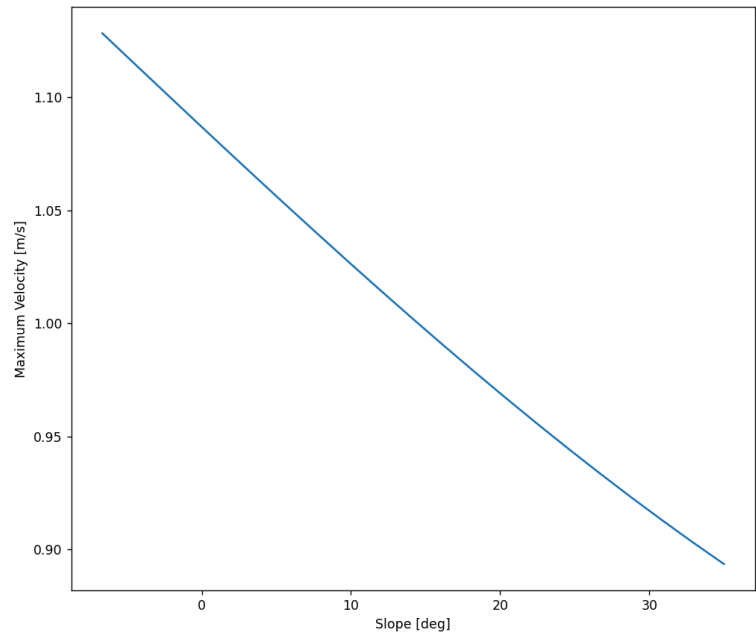
3. The motor has an 161.5 W max power at 1.9 rad/s. This is shown in the graph on the right.



4. The speed reducer significantly increases the peak power output of the power output of the reducer + motor system by approximately 3.06 times. This corresponds to the gear ratio of the system.

Rover Behavior

5. This graph shows the roughly linear ($\sin x \approx x$) effect of slope on the maximum velocity of the rover. This makes sense physically since higher slopes mean the force of gravity has a greater effect, resisting the motion of the rover and decreasing its velocity. The linearity makes sense since $\sin(x)$ is approximately linear at low angles (by Taylor series).



6.

This graph shows the linear effect of rolling resistance on the maximum velocity of the rover. The linearity of this graph makes sense because the terrain angle is kept constant and the rolling resistance acts as a linear factor when computing F_{rolling} .

This makes sense physically because the power loss due to rolling resistance is directly proportional to the rolling resistance experienced by a system.

7. The graph below shows the effect of both slope and the coefficient of rolling resistance on the rover's maximum velocity. This graph shows that the operating conditions of the rover are actually quite versatile, with only small changes in maximum velocity when the slope and coefficient of rolling resistance are changed. This can be explained by the large torque supplied by the motors, amplified by the speed reducers. Based on this graph, terrain slope is the dominant factor in the rover's maximum velocity. The effect of changing the slope is greater than changing the rolling resistance coefficient (the gradient is steeper in the direction of slope when considering this range of operating conditions).

